Detecting Medicaid Fraud with Elastic

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Medicaid Provider Spending Dataset

- Provider-level Medicaid spending data from T-MSIS, aggregated by billing or servicing provider, procedure code, and month. Covers fee-for-service, managed care, and CHIP claims from 2018-2024.
- Link

elastic

Deactivated NPIs Still Billing

elastic | Find apps, content, and more. | AI Agent | E

Discover | Check out context-aware Discover | Switch to classic | Inspect | Alerts | Save

Untitled

ES|QL help | ~ 10 years ... → now | Refresh

```
1  FROM fraud-medicaid-spending
2  | WHERE NPI_Deactivation_Date IS NOT NULL  AND CLAIM_FROM_MONTH > NPI_Deactivation_Date
3  | STATS
4  claim_months = COUNT(*),
5  total_claims = SUM(TOTAL_CLAIMS),
6  total_paid = SUM(TOTAL_PAID)
7  BY NPI, Provider_Legal_Business_Name, NPI_Deactivation_Date
8  | WHERE total_paid > 0
9  | SORT total_paid DESC
```

9 lines | @timestamp found | LIMIT 1000 rows | Quick search (CTRL K) | Show recent queries

Search field names | 0

Selected fields 6

- claim_months
- total_claims
- total_paid
- NPI
- Provider_Legal_Business_Name
- NPI_Deactivation_Date

Available fields 6

- claim_months
- NPI
- NPI_Deactivation_Date
- Provider_Legal_Business_Name
- total_claims
- total_paid

Feb 24, 2016 @ 12:19:39.533 - Feb 24, 2026 @ 12:19:39.533

63 results | Columns 6 | Sort fields

claim_months	total_claims	total_paid	NPI	Provider_Legal_Business_Name	NPI_Deactivation_Date
83	767,023	344,481,993.5	1649301789	EASTER SEALS MIDWEST	Jan 5, 2011 @ 19:00:00.000
145	1,912,330	292,266,721.859	1720471568	MAGIC HOME CARE, LLC	Jun 14, 2015 @ 20:00:00.000
410	27,149	172,924,511.375	1306182548	ALIVIA SPECIALTY, LLC	May 12, 2015 @ 20:00:00.000
617	488,977	151,990,580.016	1881622892	COMMUNITY MENTAL HEALTH FOR CENTRAL MICHIGAN	Nov 22, 2018 @ 19:00:00.000
70	1,437,591	79,685,445.063	1043294358	NURSES UNLIMITED INC	Feb 12, 2007 @ 19:00:00.000
243	467,789	62,053,128.469	1427063270	NEW YORK CITY HEALTH AND HOSPITALS CORPORATION	Jun 25, 2007 @ 20:00:00.000
67	621,250	48,233,191.219	1518175405	ACCENTCARE AT HOME, INC.	Apr 22, 2019 @ 20:00:00.000

elastic

Phantom Billing (billing more hours than are available in a day)

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Untitled

ES|QL help | Last 10 years | Refresh

```
FROM fraud-medicaid-spending
| WHERE TOTAL_UNIQUE_BENEFICIARIES > 0
| EVAL claims_per_patient = TOTAL_CLAIMS / TOTAL_UNIQUE_BENEFICIARIES
| STATS
    avg_claims_per_patient = AVG(claims_per_patient),
    max_claims_per_patient = MAX(claims_per_patient),
    p95_claims_per_patient = PERCENTILE(claims_per_patient, 95)
  BY Healthcare_Provider_Taxonomy_Code_1
| WHERE avg_claims_per_patient > 0
| SORT avg_claims_per_patient DESC
| LIMIT 20
```

11 lines @timestamp found LIMIT 0 rows | Quick search (CTRL K) | Show recent queries

Search field names

Selected fields 4
- avg_claims_per_patient
- max_claims_per_patient
- p95_claims_per_patient
- Healthcare_Provider_Taxonomy_Code_1

Available fields 4
- avg_claims_per_patient
- Healthcare_Provider_Taxonomy_Code_1
- max_claims_per_patient
- p95_claims_per_patient

Feb 24, 2016 @ 14:29:00.451 - Feb 24, 2026 @ 14:29:00.451

20 results | Columns 4 | Sort fields

avg_claims_per_patient	max_claims_per_patient	p95_claims_per_patient	Healthcare_Provider_Taxonomy_Code_1
91	91	91	2278H0200X
37.2	51	41.5	175L00000X
34.316	54	43.6	163WA2000X
29.24	39	36	102L00000X
27.9	29	28.55	2279G1100X
27.741	35	32	133VN1004X
25.25	35	31	226300000X

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High-Volume Outliers Within Specialty

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Claims per Beneficiary Ratio | High-Volume Outl... Within Specialty

ES|QL help | Last 10 years | Refresh

```
1  FROM fraud-medicaid-spending
2  | WHERE TOTAL_UNIQUE_BENEFICIARIES > 10
3  | EVAL claims_per_patient = TOTAL_CLAIMS / TOTAL_UNIQUE_BENEFICIARIES
4  | STATS
5      provider_total = SUM(TOTAL_CLAIMS),
6      provider_paid = SUM(TOTAL_PAID),
7      avg_cpp = AVG(claims_per_patient)
8    BY NPI, Provider_Legal_Business_Name, Healthcare_Provider_Taxonomy_Code_1
9  | WHERE avg_cpp > 50
10 | SORT avg_cpp DESC
11 | LIMIT 30
```

11 lines | @timestamp found | LIMIT 10 rows | Quick search (CTRL K) | Show recent queries

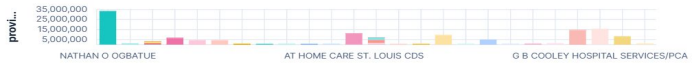
Search field names | 0

Selected fields 6

- provider_total
- provider_paid
- avg_cpp
- NPI
- Provider_Legal_Business_Name
- Healthcare_Provider_Taxonomy_Code_1

Available fields 6

- avg_cpp
- Healthcare_Provider_Taxonomy_Code_1
- NPI
- Provider_Legal_Business_Name
- provider_paid
- provider_total



Feb 24, 2016 @ 14:33:50.038 - Feb 24, 2026 @ 14:33:50.038

30 results | Columns 6 | Sort fields

provider_total	provider_paid	avg_cpp	NPI	Provider_Legal_Business_Name	Healthcare_Provider_Taxonomy_Code_1
439,688	32,397,685	263.484	1922477975	NATHAN O OGBATUE	251E00000X
31,697	527,698.195	166.25	1851557086	CHILDREN'S HOME ASSOCIATION OF ILLINOIS	320600000X
3,976	142,545.703	159	1811289473	ADAPT OF ILLINOIS, INC.	261QM0801X
3,303	143,574.109	157	1366734923	ADAPT OF ILLINOIS, INC.	261QM0801X
191,667	5,962,738.883	153.256	1457621633	JARC	253Z00000X
7,176	3,726,675	149.25	1982019816	TAKITA S. HAYES DBA TAKITA'S HELPING HANDS	253Z00000X
21,644	1,119,469.828	126	1013209675	ADAPT OF ILLINOIS, INC.	261QM0801X

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Monthly Volume Spikes

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Claims per Beneficiary Ratio | High-Volume Outl... Within Specialty | Monthly Volume Spikes

ES|QL help | Last 10 years | Refresh

```
1  FROM fraud-medicaid-spending
2  | STATS
3      monthly_paid = SUM(TOTAL_PAID),
4      monthly_claims = SUM(TOTAL_CLAIMS)
5    BY NPI, Provider_Legal_Business_Name, CLAIM_FROM_MONTH
6  | SORT NPI ASC, CLAIM_FROM_MONTH ASC
7  | STATS
8      avg_monthly_paid = AVG(monthly_paid),
9      max_monthly_paid = MAX(monthly_paid),
10     months_active = COUNT(*)
11   BY NPI, Provider_Legal_Business_Name
12 | EVAL spike_ratio = max_monthly_paid / avg_monthly_paid
13 | WHERE months_active >= 3 AND spike_ratio > 5
14 | SORT spike_ratio DESC
15 | LIMIT 25
```

15 lines @timestamp found LIMIT 3 rows | Quick search (CTRL K) | Show recent queries

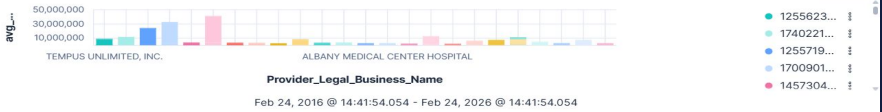
Search field names | 0

Selected fields 6

- avg_monthly_paid
- max_monthly_paid
- months_active
- NPI
- Provider_Legal_Business_Name
- spike_ratio

Available fields 6

- avg_monthly_paid
- max_monthly_paid
- months_active
- NPI
- Provider_Legal_Business_Name
- spike_ratio



25 results | Columns 6 | Sort fields

avg_monthly_paid	max_monthly_paid	months_active	NPI	Provider_Legal_Business_Name	spike_ratio
309,239.835	7,740,423.328	84	1255623716	TEMPUS UNLIMITED, INC.	25.03
514,165.912	10,281,817.75	67	1740221621	LIBERTY DIALYSIS - HAWAII LLC	19.997
1,962,965.593	21,458,712.641	84	1255719514	MARKS HOMECARE CDPAP, LLC	10.932
2,974,288.267	28,823,096.375	83	1700901725	NEW ENGLAND EMERGENCY RESPONSE SYSTEMS INC	9.691
274,314.769	2,633,522.453	30	1457304461	PAL-MED, LLC	9.6
3,947,184.05	36,374,205.719	82	1073854451	NEW JERSEY DIVISION	9.215

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Unbundling & Shell Practices

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Claims per Beneficiary Ratio | High-Volume Outl... Within Specialty | Monthly Volume Spikes | Mailing vs. Prac...ddress Divergence

ES|QL help | Last 10 years | Refresh

```
1  FROM fraud-medicaid-spending
2  | WHERE Provider_Business_Mailing_Address_State_Name IS NOT NULL
3    AND Provider_Business_Practice_Location_Address_State_Name IS NOT NULL
4  | WHERE Provider_Business_Mailing_Address_State_Name !=
5          Provider_Business_Practice_Location_Address_State_Name
6  | STATS
7      total_paid = SUM(TOTAL_PAID),
8      total_claims = SUM(TOTAL_CLAIMS),
9      months_billing = COUNT_DISTINCT(CLAIM_FROM_MONTH)
10   BY NPI, Provider_Legal_Business_Name,
11         Provider_Business_Mailing_Address_State_Name,
12         Provider_Business_Practice_Location_Address_State_Name
13 | SORT total_paid DESC
14 | LIMIT 30
```

14 lines @timestamp found LIMIT 30 rows | Quick search (CTRL K) | Show recent queries

Search field names | 0

Selected fields 7

- total_paid
- total_claims
- months_billing
- NPI
- Provider_Legal_Business_Name
- Provider_Business_Mailing_Address_State_Name
- Provider_Business_Practice_Location_Address_State_Name

Available fields 7

- months_billing
- NPI
- Provider_Business_Mailing_Address_State_Name
- Provider_Business_Practice_Location_Address_State_Name
- Provider_Legal_Business_Name
- total_claims
- total_paid

1518049...
1053321...
1538371...
1336117...
1497933...

Feb 24, 2016 @ 15:03:32.227 - Feb 24, 2026 @ 15:03:32.227

30 results | Columns 7 | Sort fields

total_paid	total_claims	months_billing	NPI	Provider_Legal_Business_Name	Provider_Business_Mailing_Address_State_Name	Provider_Business_Practice_Location_Address_State_Name
1,022,481,199.063	5,691,357	84	1518049485	HEALTH ACQUISITION CORP.	TX	NY
845,345,135.227	14,683,229	84	1053321869	GIRLING HEALTH CARE, INC.	IL	TX
527,558,826.391	2,483,630	84	1538371778	WHITE GLOVE COMMUNITY CARE, INC.	NY	NJ
508,848,315.578	3,774,010	84	1336117621	AMBERCARE HOME HEALTH CARE CORPORATION	IL	NM
476,226,485.609	8,036,581	84	1497933162	MILLENNIUM HEALTH, LLC	TX	CA

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Provider Billing Life Cycle Detection (intense billing window — they appear, bill heavily, then stop)

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D | Discover | Check out context-aware Discover | Switch to classic | Inspect | Alerts | Save

Claims per Beneficiary Ratio | High-Volume O...hin Specialty | Monthly Volume Spikes | Mailing vs. P...ess Divergence | Provider Billi...cycle Detection

ES|QL help | Last 10 years | Refresh

```
FROM fraud-medicaid-spending
| STATS
    first_claim_month = MIN(CLAIM_FROM_MONTH),
    last_claim_month = MAX(CLAIM_FROM_MONTH),
    total_paid = SUM(TOTAL_PAID),
    total_claims = SUM(TOTAL_CLAIMS),
    active_months = COUNT_DISTINCT(CLAIM_FROM_MONTH)
  BY NPI, Provider_Legal_Business_Name, Provider_Business_Mailing_Address_City_Name, Provider_Business_Mailing_Address_State_Name,
Provider_Business_Mailing_Address_Postal_Code,
    Healthcare_Provider_Taxonomy_Code_1
| EVAL lifespan_months = DATE_DIFF("month", first_claim_month, last_claim_month)
| EVAL paid_per_active_month = total_paid / active_months
| WHERE active_months <= 6 AND total_paid > 50000
| SORT paid_per_active_month DESC
| LIMIT 30
```

14 lines | @timestamp found | LIMIT 6 rows | Quick search (CTRL K) | Show recent queries

Search field names | 0

Selected fields 13

- first_claim_month
- last_claim_month
- total_paid
- total_claims
- active_months
- NPI
- Provider_Legal_Business_Name
- Provider_Business_Mailing_Address_City_Name
- Provider_Business_Mailing_Address_State_Name
- Provider_Business_Mailing_Address_Postal_Code
- Healthcare_Provider_Taxonomy_Code_1
- lifespan_months
- paid_per_active_month

Available fields 13

- active_months
- first_claim_month
- Healthcare_Provider_Taxonomy_Cod

Feb 24, 2016 @ 16:07:33.060 - Feb 24, 2026 @ 16:07:33.060

≥30 results | Columns 13 | Sort fields

⚠ Problem with 1 cluster

This cluster had issues returning data and results might be incomplete. View details

first_claim_month	last_claim_month	total_paid	total_claims	active_months	NPI	Provider_Legal_Business_Name	Provider_Business_Mailing_Address_City_...	Provider_Business_Mailing_Address_Stat...	sine Adc
Sep 30, 2022 @ 20:00:00.000	Nov 30, 2022 @ 19:00:00.000	39,762,099.5	50,853	3	1922733641	COMMUNITY HOPE WELLNESS ...	TEMPE	AZ	852
Nov 30, 2022 @ 19:00:00.000	Jan 31, 2023 @ 19:00:00.000	35,825,567.563	88,632	3	1346974763	TUSA INTEGRATED CLINIC LLC	TEMPE	AZ	852
Aug 31, 2024 @ 20:00:00.000	Oct 31, 2024 @ 20:00:00.000	22,767,544.438	4,259	3	1740221621	LIBERTY DIALYSIS - HAWAII LLC	HONOLULU	HI	968
Nov 30, 2023 @	Nov 30, 2023 @	7,313,883	48,355	1	1962158980	LIFELINE BIOSCIENCES	DOWNERS GROVE	IL	605

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Upcoding

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Claims ...y Ratio | High-V...ecialty | Monthl... Spikes | Mailin...ergence | Provid...tection | Singl...minance | Top Bi...xonomy | High-D...tration

ES|QL help | Last 10 years | Refresh

```
1  FROM fraud-medicaid-spending
2  | STATS
3      total_paid = SUM(TOTAL_PAID),
4      total_claims = SUM(TOTAL_CLAIMS),
5      avg_payment_per_claim = AVG(TOTAL_PAID / TOTAL_CLAIMS)
6    BY NPI, Provider_First_Name, Provider_Last_Name,
7       Provider_Legal_Business_Name,
8       Healthcare_Provider_Taxonomy_Code_1,
9       HCPCS_CODE, HCPCS_Short_Description
10 | WHERE total_claims > 20
11 | EVAL payment_per_claim = total_paid / total_claims
12 | SORT payment_per_claim DESC
13 | LIMIT 50
```

13 lines | @timestamp found | LIMIT 20 rows | Quick search (CTRL K) | Show recent queries

Search field names | 0

Selected fields 11

- total_paid
- total_claims
- avg_payment_per_claim
- NPI
- Provider_Legal_Business_Name
- Healthcare_Provider_Taxonomy_Code_1
- HCPCS_CODE
- HCPCS_Short_Description
- payment_per_claim
- Provider_First_Name
- Provider_Last_Name

Available fields 9

- avg_payment_per_claim
- HCPCS_CODE
- HCPCS_Short_Description
- Healthcare_Provider_Taxonomy_Code_1
- NPI
- payment_per_claim
- Provider_Legal_Business_Name
- total_claims

Feb 24, 2016 @ 16:24:56.242 - Feb 24, 2026 @ 16:24:56.242

50 results | Columns 11 | Sort fields

total_paid	total_claims	avg_payment_per_claim	NPI	Provider_Legal_Business_Name	Healthcare_Provider_Taxonomy_Code_1	HCPCS_CODE	HCPCS_Short_Description	payment_per_claim	st_
16,461,287.375	126	131,591.593	1043309735	ACCREDO HEALTH GROUP INC	3336C0004X	J2326	Inj, nusinersen, 0.1mg	130,645.138	(nu
27,316,566.688	274	101,363.598	1073569034	ACCREDO HEALTH GROUP INC	3336C0003X	J2326	Inj, nusinersen, 0.1mg	99,695.499	(nu
80,912,859.75	953	90,512.328	1346208949	ACCREDO HEALTH GROUP INC	332B00000X	J2326	Inj, nusinersen, 0.1mg	84,903.316	(nu
1,760,784.375	21	83,846.875	1376577247	ALBANY MEDICAL CENTER ...	282N00000X	J3490	Drugs unclassified injection	83,846.875	(nu
4,736,909	82	57,767.183	1336117670	RENAL TREATMENT CENTERS-...	261QE0700X	A4657	Syringe w/wo needle	57,767.183	(nu
4,035,099.281	107	37,450.667	1306182548	ALIVIA SPECIALTY	3336S0011X	J9042	Brentuximab vedotin inj	37,711.208	(nu

elastic

Year Over Year Anomaly Detection

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Claim... Ratio | High-...cialty | Mont...Spikes | Maili...rgence | Provi...ection | Singl...inance | Top ...onomy | High-...ration | Year-...ection

ES|QL help | Last 10 years | Refresh

```
FROM fraud-medicaid-spending
| EVAL claim_year = DATE_TRUNC(1 year, CLAIM_FROM_MONTH)
| STATS
    yearly_paid = SUM(TOTAL_PAID),
    yearly_claims = SUM(TOTAL_CLAIMS)
  BY NPI, Provider_Legal_Business_Name,
     Healthcare_Provider_Taxonomy_Code_1, claim_year
| STATS
    max_year_paid = MAX(yearly_paid),
    min_year_paid = MIN(yearly_paid),
    avg_year_paid = AVG(yearly_paid),
    years_active = COUNT(*)
  BY NPI, Provider_Legal_Business_Name,
     Healthcare_Provider_Taxonomy_Code_1
| WHERE years_active >= 2
| EVAL yoy_growth_ratio = max_year_paid / avg_year_paid
| WHERE yoy_growth_ratio > 3
| SORT yoy_growth_ratio DESC
| LIMIT 30
```

19 lines @timestamp found LIMIT 2 rows | Quick search (CTRL K) | Show recent queries

Search field names | 0

Selected fields 8

- max_year_paid
- min_year_paid
- avg_year_paid
- years_active
- NPI
- Provider_Legal_Business_Name
- Healthcare_Provider_Taxonomy_Code_1
- yoy_growth_ratio

Available fields 8

- avg_year_paid
- Healthcare_Provider_Taxonomy_Code_1
- max_year_paid
- min_year_paid
- NPI
- Provider_Legal_Business_Name
- years_active

Feb 24, 2016 @ 16:33:37.089 - Feb 24, 2026 @ 16:33:37.089

30 results | Columns 8 | Sort fields

max_year_paid	min_year_paid	avg_year_paid	years_active	NPI	Provider_Legal_Business_Name	Healthcare_Provider_Taxonomy_Code_1	yoy_growth_ratio
7,699,726.609	120,488.922	1,654,586.41	6	1629269618	SALUD PARA LA GENTE	261QF0400X	4.654
13,495,705.125	284,303.359	2,923,108.129	6	1174781645	SUPERIOR SENIOR CARE	251E00000X	4.617
57,731,906.266	566,816	12,675,085.447	5	1770513145	AMERICARE CERTIFIED SPECIAL ...	251E00000X	4.555
101,871,811.383	4,333,083.094	23,555,587.113	7	1255719514	MARKS HOMECARE CDPAP, LLC	251E00000X	4.325

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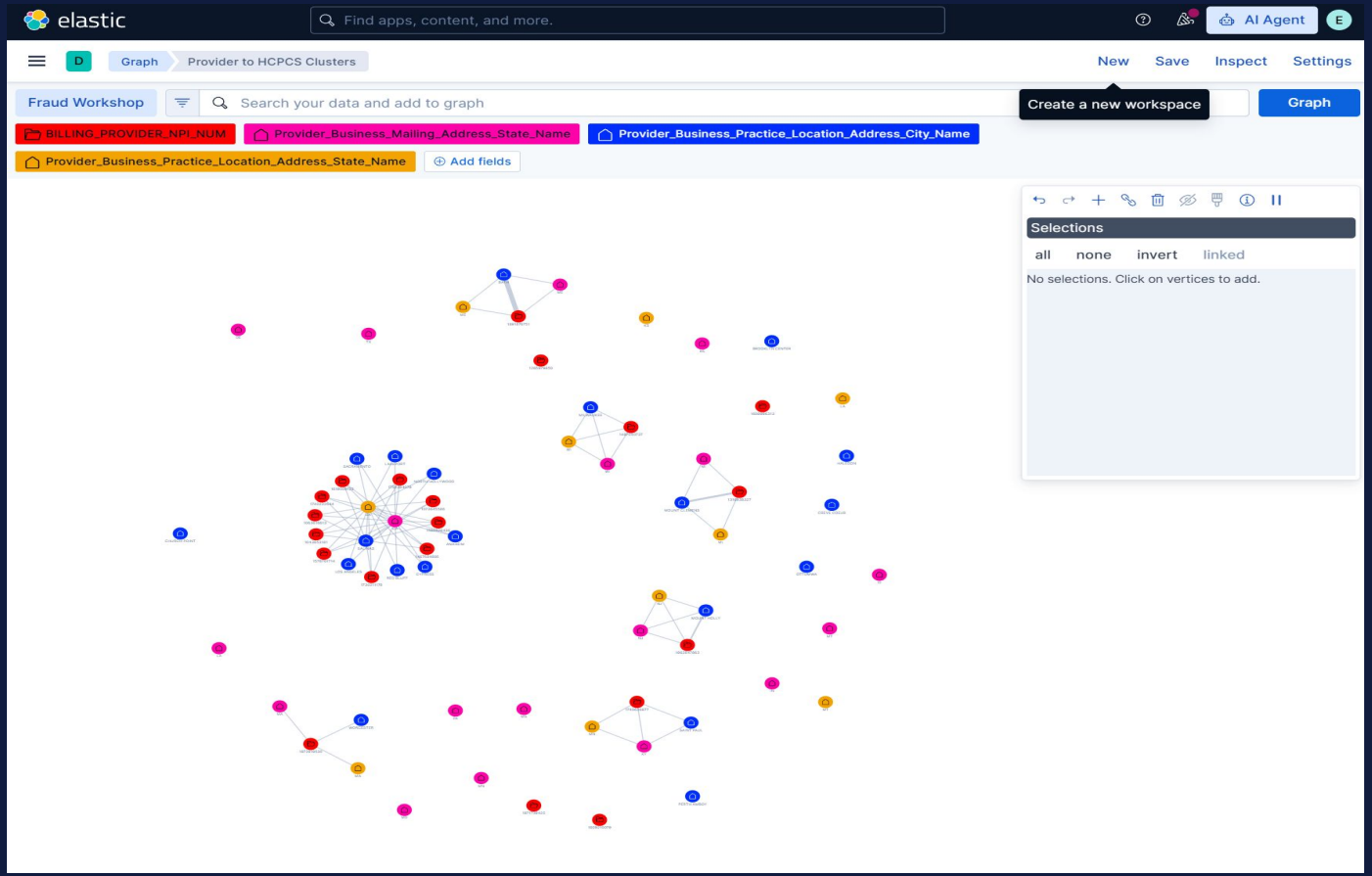
Graph to map the geographic spread of billing activity — find providers billing in states they're not licensed in, and geographic hubs where fraud concentrates.



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Anomaly Explorer

Job 1

- medicaid-provider-population-analysis is your broadest net. Four detectors on TOTAL_PAID, TOTAL_CLAIMS, TOTAL_UNIQUE_BENEFICIARIES, and mean payment — all using high_sum/high_mean over BILLING_PROVIDER_NPI_NUM. Every provider is scored against the full population baseline.

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Anomaly Explorer (2)

Job 2

- medicaid-rare-billing-patterns is the lightest job and catches a completely different fraud class — novel and infrequent events rather than volume outliers. The rare on billing/servicing NPI pairings is particularly useful for NPI harvesting, and the freq_rare on provider within taxonomy catches burst-and-disappear schemes.

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Anomaly Explorer (3)

Job 3

- medicaid-advanced-peer-behavioral-profiling is the production-grade job. The key distinction from Job 1 is the by_field_name: Healthcare_Provider_Taxonomy_Code_1 — every provider is now scored against their specialty peer group, not the general population. The low_sum on beneficiaries is also a deliberate inversion — high spend + suspiciously few patients is the phantom billing signature that a high_sum alone would miss.

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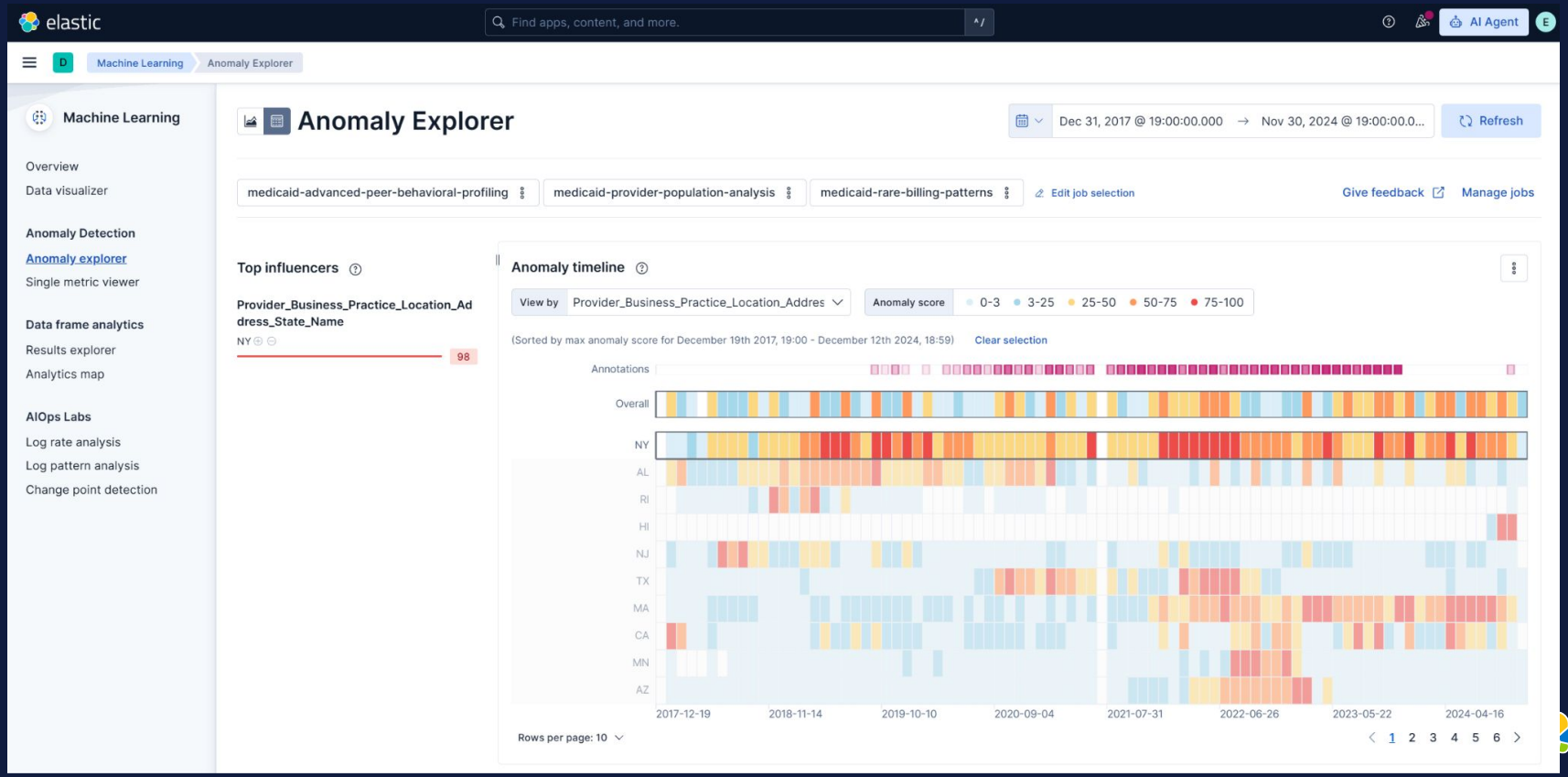
Anomaly Explorer Screenshots



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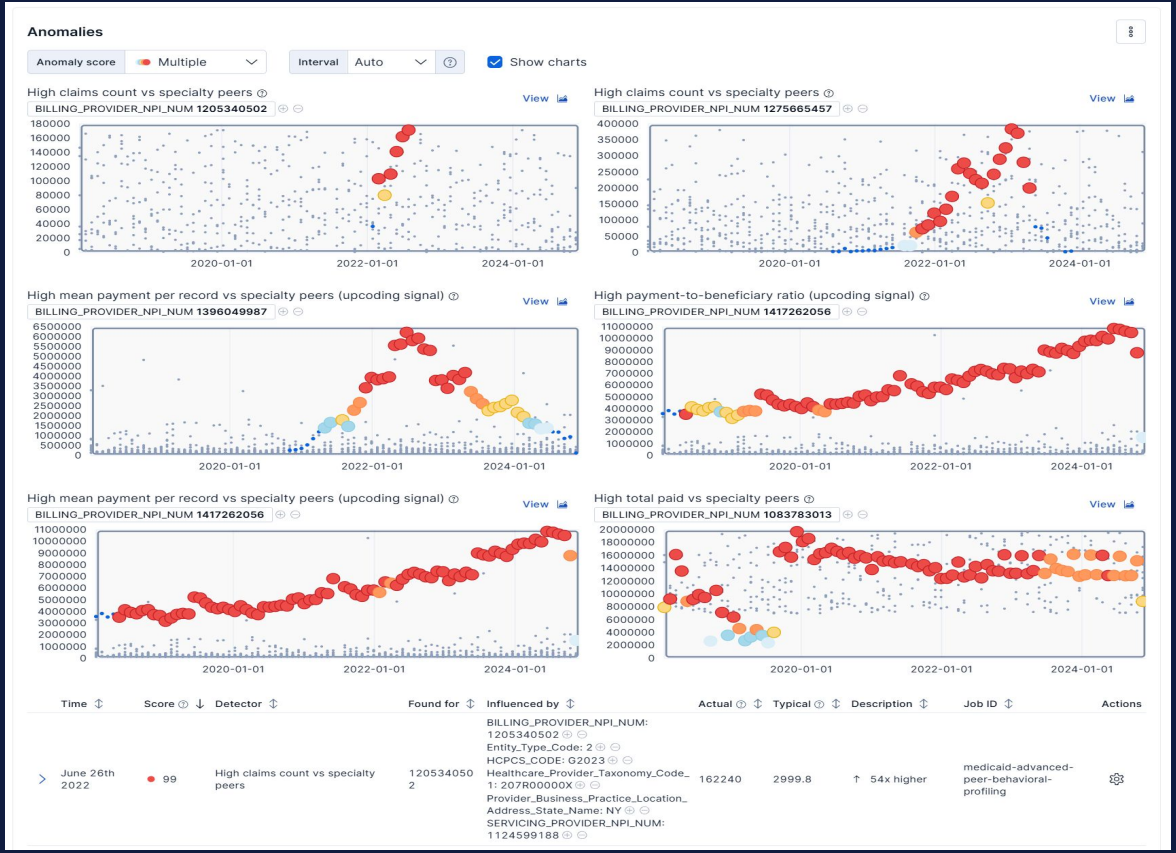
Anomaly Explorer Screenshots (2)



elastic